Nonlinear 3D Euler: checked solver foundations

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Accuracy and optimality remain unproved. Lean has checked the named foundations listed below. The result is not yet a solver certified to reach a specified physical error, and no matching total-work lower bound has been established.

The physical problem

Full, unforced, compressible Euler with $\gamma = 2$ on the periodic cube $[0, 2\pi]^3$. The family has amplitude $A \in [0.5, 1]$ and density perturbation $\delta \in [-0.05, 0.05]$. Only initial conditions are supplied; all five conserved fields must evolve.

$\rho_0 = 1 + \delta \sin x \sin y \sin z$
$v_0 = A (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0)$
$p_0 = 1 + (A^2/16)(\cos 2x + \cos 2y)(\cos 2z + 2)$
$E_0 = p_0 + \rho_0 |v_0|^2/2$

Initially $\rho_0 \geq 0.95$ and $p_0 \geq 0.625$. At $(0, 0, \pi/4)$, the vertical momentum equation forces $\partial_t(\rho v_z) = A^2/4 \geq 1/16$. This rules out freezing the initial zero vertical velocity. It does not prove that no analytic representation could ever exist.

Candidate and intended result

The baseline is a conservative, explicit, global-speed Rusanov finite-volume method. An alternative uses an existing entropy-conservative flux plus LF diffusion and a timestep-halving acceptance test, now with explicit per-state bounds (pages 7–8). Both use analytic initial averages. $T = 0.25$ is the numerical target; a common smooth existence interval is still unproved. The intended error includes the full evolved field, spatial reconstruction and time integration, using volume-normalized L1 error in each conserved component.

Required for completion	Current status
Common smooth time and uniform regularity	Unproved
Fully discrete nonlinear stability and continuum error	Unproved; foundations checked
Scientifically justified competitor class	Initial search scope; not finalized
Matching total-work upper and lower bounds	Unproved

All competitors receive the explicit initial-condition formulas. The target is not initial-data sampling complexity, a cache schedule, a CFL count, an output-size bound, or a ranking of a finite benchmark catalogue.

What Lean has checked

The retained audit checks **69 named statements** with Lean (version 4.24.0, x86_64-unknown-linux-gnu, commit 797c613eb9b6d4ec95db23e3e00af9ac6657f24b, Release). Transitive axiom dependencies are limited to the standard principles propext, Classical.choice and Quot.sound. Statements must still be interpreted with their definitions and explicit premises.

Layer	Checked claim and its limit
Actual nonlinear finite-volume update	Exact-real positive density/pressure from computed wave speeds and a valid CFL ratio; periodic conservation of all five components. This does not prove convergence.
Conservative / primitive connection	Actual derivative rules, invertible state-map derivative, equivalent Euler residuals, and a positive symmetrizing energy. This does not construct a solution.
Frozen principal update	The constant-coefficient linearized update contracts its discrete energy under the stated wave-speed and CFL bounds.
Varying principal update	An exact energy defect and an explicit growth estimate from stated spatial difference bounds. This is not the full nonlinear update stability theorem.
Grid embedding and state-box bounds	A mesh-independent 3D embedding and explicit coercivity/coefficient bounds. The trajectory's derivative energies and state-box membership are not bounded yet.
Actual mathematical entropy	Conserved-variable gradient and positive Hessian, convexity on admissible states, finite averaging bound, canonical entropy-flux potential and local entropy balance implied by differentiable Euler balance.
Actual finite-step entropy estimate	Flux residual, six-neighbor increment bound and finite-step Taylor remainder combine into entropy dissipation under explicit convex-region and step conditions. These conditions are not established for the full numerical trajectory.
Concrete bounds and obstruction	A convex physical region contains the full initial family and has proved curvature/flux bounds. An exact witness shows the present global viscosity criterion cannot certify the candidate's wave-speed recipe there.

Principal-update energy estimate

Write $q = 1/\rho$, $e = (e_q, e_v, e_p)$ and $g_{q,p}(e,e) = e_p^2/(2p) + |e_v|^2/q + (e_q/q + e_p/(2p))^2$. Sum this over cells to obtain E. For the principal update, the proof gives

$$\mathbf{E_{after} \le [1 + 756\,\Delta t\,(256\,B\,K + 32\,D)]\,E_{before}}.$$

Both energies use the old local metric. Premises include $1/4 \le q,p \le 4$, $|a_d| + |v_d| \le B$, a valid CFL ratio, an identity central shift, and neighbor differences satisfying $|\Delta q| + |\Delta p| \le hK$ and $|\Delta v_d| \le hD$. Here $\Delta t = \text{ratio} \times h$. The constant does not depend on the number of cells when B, K and D have uniform bounds. Proving such bounds, the time variation of the metric and the nonlinear remainder are outstanding steps.

Exact negative control: with two cells, $q = (1,2)$, $p = 1$, $v = 0$, directional speeds 2 and ratio 1/12, the CFL number is 1/2. A specific error field has energy 5 before and 125/16 = 7.8125 after the principal update. Thus a varying local metric need not contract. This error-field example does not replace the nonlinear Taylor–Green research family.

Grid control and numerical evidence

The checked embedding gives $|f(i,j,k)|^2 \le 8\, E_3(f)$. In one dimension, $E_1(f) = n^{-1} \Sigma f^2 + n \Sigma(f_{k+1} - f_k)^2$; E_2 and E_3 apply this construction recursively in the other directions. It contains the eight mixed differences with each coordinate differentiated at most once, scaled to the unit cube. The constant 8 is independent of n. This controls values from derivative energies; it does not bound those energies along the Euler evolution.

Independent candidate checks

Initial cell averages agree with independent eight-point-per-axis Gauss quadrature to 9.99e-16 in the tested cells. A randomized admissible-field update has conservation drift 3.55e-15. An intentionally too-small split speed gives pressure −49, independently reproduced by Lean. These checks do not certify floating-point error or continuum accuracy.

n	A / δ	Steps	min ρ	min p	max |v_z|
16	0.5 / -0.05	11	0.964880	0.971706	0.008675
32	0.5 / -0.05	23	0.957623	0.944254	0.012397
16	0.75 / 0	12	0.997845	0.944951	0.018992
32	0.75 / 0	25	0.997943	0.882108	0.027463
16	1 / 0.05	13	0.969251	0.916210	0.032740
32	1 / 0.05	27	0.961357	0.803544	0.047906

All six runs end at T = 0.25. They show evolving vertical velocity and admissible sampled states on two grids. Their success neither proves the full parameter family remains smooth nor supplies an exact continuum reference. No speedup or optimality claim is inferred from these runs.

Reproducibility and next proof obligations

The source is under demos/lean_euler/nonlinear/formal. record/solver contains the frozen assembled Lean input, generated audit, logs, result JSON, component hashes, numerical checks and local-lab job receipts. The Python method is not a verified extraction of the Lean definitions.

Run from the repository root with a new output directory:
.venv/bin/python demos/lean_euler/nonlinear/run_solver_study.py
--output artifacts/nonlinear-energy-replay

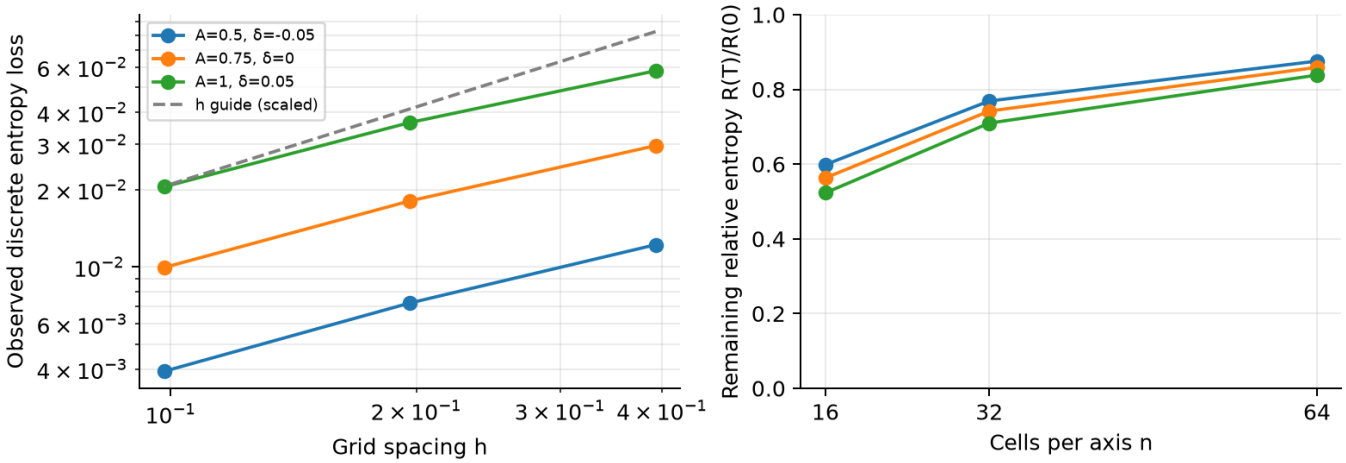
The next required bridge is quantitative regularity and a stability/error estimate for the actual nonlinear update. Initialization and continuous reconstruction must then be connected to the evolved PDE solution. A complete cost model and a matching necessary-accuracy lower bound remain essential before the goal can be completed.

Assembled proof SHA-256: 5de26c3d6b5c2b51d7df4c3b6619c768
6032c506196c55ec4bdec8ed6b005a97
Draft PR: github.com/tomzhu0225/simjecture/pull/90

Entropy: the proposed error mechanism

For γ = 2, the mathematical entropy is η(U) = −ρ log(p/ρ²). Lean now checks its actual gradient and Hessian, convexity on the admissible conserved-state set, and the local entropy balance that follows from differentiable Euler balance. The Hessian was derived from η, not defined as a convenient positive form. Existence of a smooth solution and the periodic integral conservation theorem are still required.

A further checked result says a finite averaging map with nonnegative weights and unit row/column sums cannot increase this convex entropy. A bridge to the actual continuous output reconstruction remains unfinished.



Observed quantities, not certified continuum errors. Nine runs of the actual Rusanov update use n = 16, 32, 64, CFL = 0.3 and T = 0.25. Entropy loss is measured between discrete initial and final states. R is relative entropy about the conserved grid mean. The scaled h line is a visual guide; no rate or proof constant is inferred from this finite sample.

A / δ (n = 64)	Entropy loss / h	R(T) / R(0)	Min observed drop / (Δt h R)
0.5 / -0.05	0.039945	0.875728	5.368747
0.75 / 0	0.101323	0.859054	6.125759
1 / 0.05	0.208938	0.838228	7.076122

Remaining argument. Develop a sharper finite-step estimate compatible with the candidate, and prove the uniform trajectory conditions. Connect accumulated loss to final continuum error through proved state bounds and the reconstruction. Only then could a necessary mesh resolution support a total-work lower bound. The data above do not discharge any of these obligations.

The initial search scope includes higher-order methods; first-order Rusanov cannot be declared optimal in that scope without further argument. The proposed LF/Rusanov class with convex reconstruction is still under investigation, with explicit initial formulas and all preprocessing available to competitors.

Reference: Tadmor, Acta Numerica 12 (2003), §§7–8, on the balance between spatial entropy dissipation and explicit time discretization. The reference is not imported as a Lean theorem.

A checked finite-step theorem

The new theorem concerns the actual nonlinear Rusanov finite-volume update. It includes explicit time discretization. It is a conditional method theorem; the required uniform trajectory membership and a useful certified parameter choice remain to be established. Concrete region constants and their limitation are checked on the following page.

The region conditions

Let K be a convex set of positive-density, positive-pressure conserved states. Both the old and the updated cell states must lie in K. Throughout K, for every conserved-state tangent k, require
$\lambda\,|k|^2 \le D^2\eta(U)[k,k] \le M\,|k|^2$,
and $|F_d(V) - F_d(U)|^2 \le L^2\,|V - U|^2$ for every direction. These are bounds on the actual physical entropy Hessian and flux, using the sum of the five squared nondimensional components. They are not numerical error estimates supplied as assumptions.

The viscosity and timestep conditions

Here viscosity refers to artificial diffusion in the numerical face flux: the Rusanov term is $-(a_d/2)(U_{right} - U_{left})$. It is not physical fluid viscosity. The target PDE remains inviscid 3D Euler.

Write $r = \Delta t/h$ and $0 < a_{min} \le a_d \le a_{max}$ for the directional global viscosities. A sufficient pair of conditions is
$M(1 + L^2) \le a_{min}\lambda/2$,
$3\,M\,r\,(a_{max}^{\,2} + L^2) \le a_{min}\lambda/8$.

With $H_{total} = \Sigma_{cells}\,\eta(U)$ and $J = \Sigma_{forward\ periodic\ faces}\,|U_{right} - U_{left}|^2$, Lean proves
H_{total}(after) $\le$ H_{total}(before) $-$ (r a_{min} λ/8) J.

How the proof connects to the update

Checked piece	Role
Actual entropy calculus	The derivative of the actual entropy gradient pairs with the Euler flux to give the momentum component of the tangent.
Rusanov face residual	The Hessian and flux bounds control the central flux term; sufficiently large LF viscosity supplies a negative quadratic face contribution.
Periodic sum and finite step	The entropy potential telescopes. A six-neighbor quadratic bound and an actual Hessian-based Taylor estimate control temporal entropy production.

Practical limitation. On the concrete region considered next, this sufficient viscosity criterion cannot certify the current wave-speed recipe, even with improved global curvature constants. A smaller timestep alone cannot resolve that viscosity condition. This does not refute the solver's actual entropy behaviour. Uniform trajectory bounds, continuum error and work optimality remain unproved.

Concrete bounds and a proof-method obstruction

Lean now checks a concrete convex region K of conserved states with $1/4 \le \rho \le 4$, total energy density $E \le 4$, and $p \ge 1/4$. It implies $p \le 4$ and $|v|^2 \le 30$. The complete initial family belongs to K for every allowed A, δ and every spatial point. This is initial membership, not invariance during the nonlinear evolution.

Explicit bounds, derived from the physical definitions

Throughout K, the checked constants are $\lambda = 1/12500$, $M = 100000$ and $L = 1200$. The Hessian is the actual conserved-variable entropy Hessian. The finite flux bound follows from proved physical-flux derivatives and the mean value theorem. The constants are deliberately coarse. Inserting them into the preceding viscosity criterion would require $a \ge 3{,}600{,}002{,}500{,}000{,}000$, which is not a useful certificate for the present method.

Improving those constants alone is insufficient on K

The rest state $U^* = (1,0,0,0,1)$ belongs to K and occurs in every member of the actual initial family at $(0, \pi/2, \pi/2)$. Lean checks two exact directional Hessian values there:

Direction k	$\lvert k\rvert^2$	$D^2\eta(U^*)[k,k]$	Necessary bound
$(1,0,0,0,2)$	5	2	$\lambda \le 2/5$
$(2,0,0,0,-1)$	5	13	$M \ge 13/5$

For every state in K, each physical directional wave speed is at most 12. Therefore the candidate's computed global directional speeds are at most 12 whenever its grid states are in K. The sufficient condition then has a left side $M(1+L^2) \ge 13/5$ and a right side $a\lambda/2 \le 12/5$. It is impossible. This obstruction holds for any valid global curvature constants on K, not only the coarse numerical values above.

Meaning. This rules out certifying the candidate with this particular global sufficient criterion on K. It does not prove the numerical method unstable, falsify the observed entropy loss, or show that Euler solver optimality is impossible. The physical problem and accuracy goal are unchanged.

Next direction. The alternative flux on the following page cancels the central face entropy residual exactly, avoiding this criterion. Uniform regularity, common smooth existence to $T = 0.25$, continuous reconstruction, full nonlinear accuracy and matching work bounds still require proof.

An alternative with a terminating one-step search

The new candidate uses Chandrashekar's existing logarithmic-mean entropy-conservative flux, plus $-(a_d/2)(U_R-U_L)$. It changes the numerical method; no exact equivalence to the Rusanov baseline or speedup is claimed. The physical PDE remains inviscid Euler.

What the new Lean proof establishes

The actual five-component flux satisfies the physical Euler consistency and canonical entropy jump identities in all three directions. Its dissipative face residual is nonpositive for **any nonnegative a_d**. This eliminates the large-diffusion condition of the previous face estimate.

For any finite admissible periodic grid, define D as the sum of $(a_d/2)\,\Delta\eta' \cdot \Delta U$ over forward faces, and $H = \Sigma\eta(U)$. Freeze the old-state residual while halving a positive trial ratio $r = \Delta t/h$. Accept a trial only when every cell has positive density and pressure and
$H(U_{trial}) \le H(U) - rD/2$.

Lean proves that the derivative of trial entropy at $r = 0$ is $-D$. For $D > 0$, sufficiently small positive trials pass. For $D = 0$ with positive coefficients, strict entropy-gradient monotonicity forces equal states across neighboring faces; the residual is then zero. Thus **a finite real-arithmetic halving search succeeds**. The first accepted positive step preserves all conserved totals, admissibility and the finite-step entropy inequality; earlier trials fail its acceptance test.

Independent numerical checks

Logarithmic means are compared with 70-digit arithmetic. An independent complex-step entropy gradient checks 1,024 random state pairs in all directions. The maximum scaled entropy-identity residual is 2.09e-14. An excessive proposal triggers two halvings; a constant-state control stays unchanged.

A / δ at n = 64	Accepted steps	Rejected	Min ρ / min p
0.5 / -0.05	45	0	0.95031 / 0.90775
0.75 / 0	50	0	0.99782 / 0.79245
1 / 0.05	55	0	0.95031 / 0.63103

All nine $16^3/32^3/64^3$ runs reach T = 0.25 with ordinary wave-speed diffusion and CFL = 0.3, without rejected trials. All sampled entropy margins are nonnegative. These are finite observations, not universal accuracy or cost bounds.

Remaining gap. The theorem uses exact real comparisons, not verified floating-point guards. It gives no uniform retry/timestep bound, no guarantee for Python's cap of 60 halvings, and no proof of reaching T in finitely many steps. Continuum accuracy, complete computational work and a matching competitor lower bound remain unproved. Page 8 adds explicit per-state step/retry bounds; uniform trajectory control remains open.

Reference: Chandrashekar (2013), section 4.4. Sources: LogMean.lean, EntropyConservativeFlux.lean, EntropyCandidateStep.lean. Retained evidence: record/solver, including ec-validation.json and the 69-statement audit.

Quantitative step and retry bounds

The new proof bounds the accepted timestep and the number of halvings from the current grid state. It uses the actual directional entropy curvature, avoiding the earlier coarse global Hessian constant. The numerical method is unchanged.

Bounds derived from the conserved variables

For one cell, write $U = (\rho, m, E)$ and let k be its frozen update residual. Define
$Q = 2\rho E - |m|^2 = 2\rho p$,
$Q_1 = 2k_\rho E + 2\rho k_E - 2m \cdot k_m$,
$Q_2 = 2k_\rho k_E - |k_m|^2$, and $W = |Q_1| + 2|Q_2|$.

Along $U + sk$, the pressure polynomial is exactly $Q + sQ_1 + s^2Q_2$. With
$R = 1 / \{2[1 + \max_{cells}(|k_\rho|/\rho + (|Q_1|+|Q_2|)/Q)]\}$,
Lean proves positive density/pressure for every $0 \le s \le R$. The actual entropy second directional derivative is at most the explicit cell value
$b = 6k_\rho^2/\rho + 4|k_\rho|W/Q + 6\rho|Q_2|/Q + 6\rho(W/Q)^2$.

The resulting quantitative guarantee

Let $B = \Sigma b$ and let D be the face entropy dissipation from page 7. For $D > 0$, set $c = \min(R, D/(B+D))$. Lean proves $c > 0$ and
$H(U+rk) \le H(U) - rD + r^2B/2$ for $0 \le r \le R$.
Every $0 \le r \le c$ therefore passes the positivity/entropy acceptance test.

For a positive proposed ratio r_0, the first accepted halving index is at most **$N = \text{clog}_2(\text{ceil}(r_0/c))$**, where $\text{clog}_2(n)$ is the least integer $N \ge 0$ with $n \le 2^N$. The accepted ratio is at least **$\min(r_0, c/2)$**. When $D = 0$, the stationary update accepts the original proposal with zero halvings.

Numerical evaluation of the bound

Cells per axis	Largest proposed r_0/c	Predicted / actual halvings
16	0.2527	0 / 0
32	0.1182	0 / 0
64	0.1150	0 / 0

These nine trajectories use ordinary wave speeds, CFL = 0.3 and T = 0.25. An excessive proposal has a bound of six halvings and actually uses two. Independent complex-step Hessians on 2,048 random states and five trial positions agree with the polynomial expression to 1.16e-13 maximum scaled error. The solver does not execute this diagnostic bound.

What remains. The cap is proved from each current state. There is no proved uniform lower bound for it over the entire evolving parameter family and all meshes. Floating-point evaluations are not certified enclosures. Target-time completion, continuum accuracy, complete computational work and matching optimality remain open.

Sources: PressurePolynomial.lean and EntropyStepBounds.lean. Five bounded jobs replayed that checkpoint; step-bound-validation.json retains the measurements and their source hashes. The current six-job replay adds the next page's checks.

Regional estimates and an invariance counterexample

A uniform cost argument needs control of the evolving states. Lean now proves a regional flux estimate and checks why positivity, conservation and entropy decrease alone do not establish the required region membership.

The actual alternative flux on the physical region

For $K = \{1/4 \le \rho \le 4, E \le 4, p \ge 1/4\}$, the checked Euclidean estimate is
$|f^{EC}_d(U,V) - F_d(U)| \le 3 \cdot 10^8 |V-U|$,
with the same bound relative to $F_d(V)$. It follows from the physical state bounds and logarithmic-mean interval property. The large constant is a coarse analytical bound; the solver continues to use computed wave speeds. This is an ingredient for a residual estimate, not yet nonlinear stability or continuum accuracy.

A checked failure of global region invariance

On a periodic 3^3 grid, set $\rho = 1$ and $E = 4$ everywhere. Let x-momentum be $(1,0,-1)$ on successive x-planes, with the other momenta zero. All initial states belong to K. At the middle plane, Lean proves the exact residual
$k_E = 1/[2\ LM(1/7,1/8)] + 56/15 > 0$.
Consequently every positive ratio r gives $E_{new} = 4 + rk_E > 4$, independently of the diffusion coefficients. The theorem applies to the actual first entropy-accepted step.

Compression control	Observed value
Old / new maximum cell energy	4 / 4.240972
Entropy acceptance margin	0.013127
Rejected trials / conservation drift	0 / 2.22e-16

The cell energy ceiling fails while density/pressure stay positive, the entropy test passes and total energy is conserved. Adding $E \le 4$ as an extra acceptance guard would reject every positive trial for these data.

Implication for the target family

The compression data are not the Taylor–Green family. This counterexample rejects a global method-level invariance shortcut. It does not refute a family-specific or time-dependent state bound. Those bounds still require proof.

Target-family grid	Parameter cases	Smallest sampled margin inside K
16^3	25	0.43461
32^3	3	0.39075
64^3	3	0.37900

The 31 target-family trajectories stay in K numerically through $T = 0.25$. The margin is the minimum of $\rho-1/4$, $4-\rho$, $4-E$ and $p-1/4$ over sampled states. The 16^3 scan covers five amplitudes and five density perturbations. These finite float observations do not discharge a universal evolution theorem.

Sources: ECFluxBounds.lean, RegionEscape.lean and region-validation.json. The full accuracy-and-work-optimality goal remains open.